Skills Summary

Composite Solids, Surface Area, and Density Design

Use this reference while completing your worksheet or when studying.

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1. Volume of a composite solid (composite-volume)

Volumes add. Cut the solid at the joins, find each piece, add. $V_{\text{cyl}} = \pi r^2 h$, $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$, $V_{\text{sphere}} = \frac{4}{3}\pi r^3$ (hemisphere: $\frac{2}{3}\pi r^3$), $V_{\text{prism}} = Bh$.

A *hole* is a piece you **subtract**: $V = V_{\text{solid}} - V_{\text{hole}}$. Keep π symbolic until the last line so nothing is lost to rounding.

Example: A tank is a cylinder ($r = 2$, $h = 6$ cm) with a hemisphere lid of the same radius. Find its volume.
Step 1. Two solids joined at a shared circle, and volume asks how much space is inside, so the two volumes simply add — the shared circle has no volume.
Step 2. $\pi(2)^2(6) = 24\pi$ and $\frac{2}{3}\pi(2)^3 = \frac{16}{3}\pi$, so $V = \frac{88}{3}\pi = \mathbf{92.15}$ $\mathbf{cm^3}$.

Try it: Cylinder $r = 3$, $h = 5$ cm with a hemisphere lid. Find the volume.

check: **197.92** cm^3

2. Surface area of a composite solid

Surfaces do NOT add. Count only the faces you could paint: at every join, both pieces lose the circle (or face) they share.

Cylinder side $= 2\pi rh$, cone slant side $= \pi r\ell$, circle $= \pi r^2$, sphere $= 4\pi r^2$ (hemisphere curved part: $2\pi r^2$).

A drilled hole *removes* two circles and *adds* the wall $2\pi rh$ inside.

Example: Same tank: cylinder $r = 2$, $h = 6$ cm, hemisphere lid. Find the outside surface area.
Step 1. Walk the outside of the solid: the flat base, the cylinder wall, the dome — the circle where the dome meets the cylinder is inside, so it is not counted.
Step 2. $\pi(2)^2 + 2\pi(2)(6) + 2\pi(2)^2 = 4\pi + 24\pi + 8\pi = 36\pi = \mathbf{113.10}$ $\mathbf{cm^2}$.

Try it: Cylinder $r = 3$, $h = 5$ cm with a hemisphere lid: outside surface area (base + wall + dome).

check: **179.07** cm^2

3. Density: turning volume into mass (density-mass)

$$\text{density} = \frac{\text{mass}}{\text{volume}} \qquad \Longrightarrow \qquad m = \rho V \qquad \text{and} \qquad V = \frac{m}{\rho}$$

Units decide the arithmetic: a density in g/cm^3 needs a volume in cm^3; a density in kg/m^3 needs m^3. Find the volume first, then multiply — never multiply density by a length.

Example: A solid aluminium cylinder, $r = 2$ cm, $h = 8$ cm, density 2.7 g/cm^3. Find its mass.
Step 1. Mass is asked and density is given, so the missing piece is volume — compute V, then multiply by the density.
Step 2. $V = \pi(2)^2(8) = 32\pi$ cm^3, so $m = 2.7 \times 32\pi = 86.4\pi =$ **271.43** g.

Try it: A solid copper sphere, $r = 2$ cm, density 8.96 g/cm^3. Find its mass.

check: **300.25** g

4. Design backwards: solve for a dimension

A design problem fixes the *answer* (a volume, a mass) and asks for a *dimension*. Write the formula as an equation, then solve:

$$\pi r^2 h = V \Rightarrow h = \frac{V}{\pi r^2}, \qquad \tfrac{4}{3}\pi r^3 = V \Rightarrow r = \sqrt[3]{\frac{3V}{4\pi}}, \qquad V = \frac{m}{\rho} \text{ first if a mass is given.}$$

Example: A cylindrical can of radius 5 cm must hold 400 cm^3. How tall must it be?
Step 1. The volume is given and a dimension is unknown, so set the volume formula equal to the target and solve for h instead of computing forward.
Step 2. $\pi(5)^2 h = 400$, so $h = \dfrac{400}{25\pi} =$ **5.09** cm.

Try it: A cylindrical can of radius 4 cm must hold 300 cm^3. Find its height.

check: **5.97** cm

(!) Watch out: Do not add whole surface areas at a join — that double-counts the hidden circle. And if a design fixes the volume, changing the radius cannot change the mass: $m = \rho V$, so a wider solid is simply a shorter one.